

Automated Feature Extraction of Distributed Wearable Ultrasound Signals During Highly Dynamic Physical Activity

Erica L. King^{1,2}, Qi Wei^{1,2}, Margaret T. Jones^{1,2,4}, Siddhartha Sikdar^{1,2,3}, Parag V. Chitnis^{1,2,3}

¹Department of Bioengineering, George Mason University, Fairfax VA, USA

²Center for MedTech Innovation, George Mason University, Fairfax VA, USA

³Myokinetics, Inc., Baltimore MD, USA

⁴Patriot Performance Lab, George Mason University, Fairfax VA, USA

Background, Motivation and Objective: Musculoskeletal ultrasound technology is increasingly utilized for assessing muscle activation in real-time. However, bulky handheld probes and operator dependency of conventional systems hinder imaging during physical activity. We have developed a compact, distributed wearable ultrasound (D-WUS) system that employs time-delayed spectrometry (TDS), a frequency-domain technique that uses chirp transmit-receive signals for multi-site M-mode imaging. This study presents an automated processing pipeline designed specifically for D-WUS data, enabling real-time muscle-level metrics essential for monitoring rehabilitation and recovery.

Statement of Contribution/Methods: A wearable 4-channel D-WUS system was used to collect M-mode data from 20 subjects performing bilateral countermovement jumps (CMJs). The WUS system transmitted 20-ms chirps spanning 3-5 MHz and acquired echo signals from 4 transducers distributed over different muscles, left and right vastus lateralis (VL) and vastus medialis oblique (VMO) muscles (Fig. A1), at a rate of 50 lines/s. Echo signals down-mixed, low-pass filtered, and Fourier transformed in real-time to obtain M-mode data (Fig. A). M-mode data was processed using a composite factorization technique utilizing a non-negative matrix factorization (NNMF) across a range of factors (1:50). The factors were then concatenated and processed using principal component analysis to obtain the primary time-domain component (Fig. C) that described tissue contraction during CMJ (Fig. B). Correlation analysis was performed on these time-domain signals from different muscles to examine muscle coordination healthy individuals and subjects with reported knee injury.

Results/Discussion: NNMF-based processing automatically provided a time trace that visually tracks the movement in tissue interfaces in M-mode images and correlated well with the bilateral force-time measurements (Fig. C). Participants with prior knee injuries showed decreased VL-VMO correlation ($p=0.039$) and increased activation time-lags ($p=0.013$). These findings indicate that intra-limb coordination deficits could be used as an indicator of persistent dysfunction post-injury. This D-WUS approach with automated NNMF processing can provide actionable metrics beneficial for real-time rehabilitation feedback.

